Program-3: Code a Dockerized Python Flask Application

Step 1: Create Project Directory

mkdir Program3
cd Program3

Step 2: Create Dockerfile

Create a file named Dockerfile

nano Dockerfile

Add the following content:

```
# Use official Python image
FROM python:latest

# Set working directory
WORKDIR /app

# Copy requirements file and install dependencies
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

# Copy application code
COPY . .

# Expose port 5000
EXPOSE 5000

# Run the application
CMD ["python", "app.py"]
```

```
  GNU nano 7.2                                                                Dockerfile
FROM python:latest

WORKDIR /app

COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

COPY . .

EXPOSE 4000

CMD ["python","app.py"]
```

Save and exit (CTRL+O, ENTER, CTRL+X).

Step 3: Create Flask Application File

nano app.py

Add:

```
from flask import Flask

app = Flask(__name__)

@app.route("/")
def home():
    return "Hello from Simple Flask Docker!"

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

```
 GNU nano 7.2                                                                          app.py

from flask import Flask

app = Flask(__name__)

@app.route("/")
def home():
    return "Hello from Simple Flask Docker!"

if __name__=="__main__":
    app.run(host="0.0.0.0",port=4000)
```

Save and exit.

Step 4: Create requirements.txt

nano requirements.txt

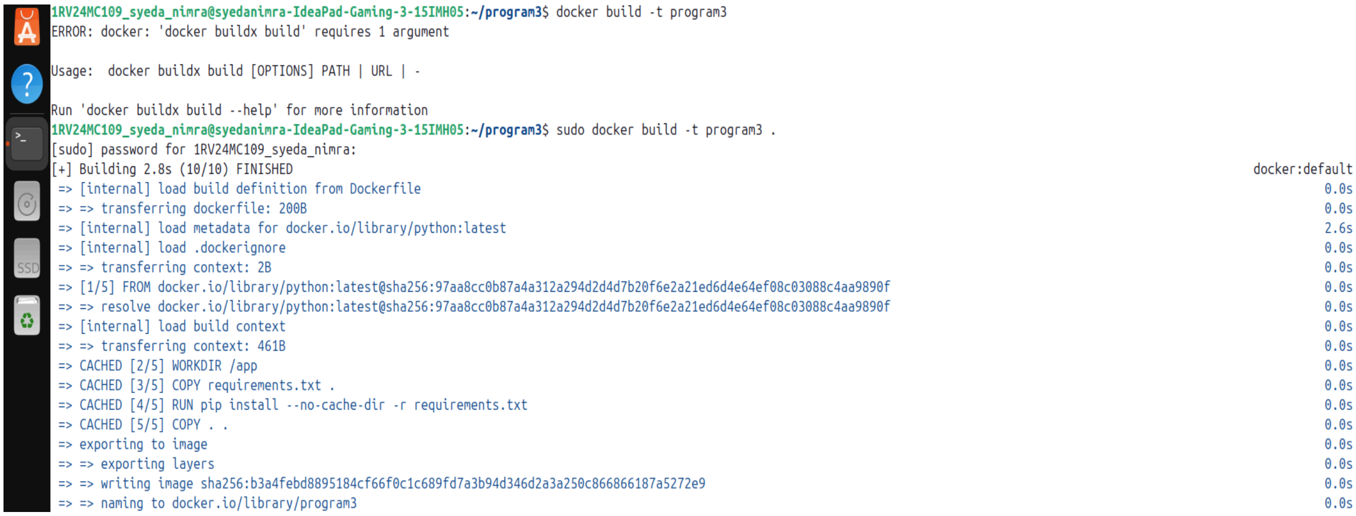
Add:

Flask==2.3.3

Save and exit.

Step 5: Build Docker Image

docker build -t program-3 .



Verify:

docker images

Step 6: Run Docker Container

docker run -p 5000:5000 program-3

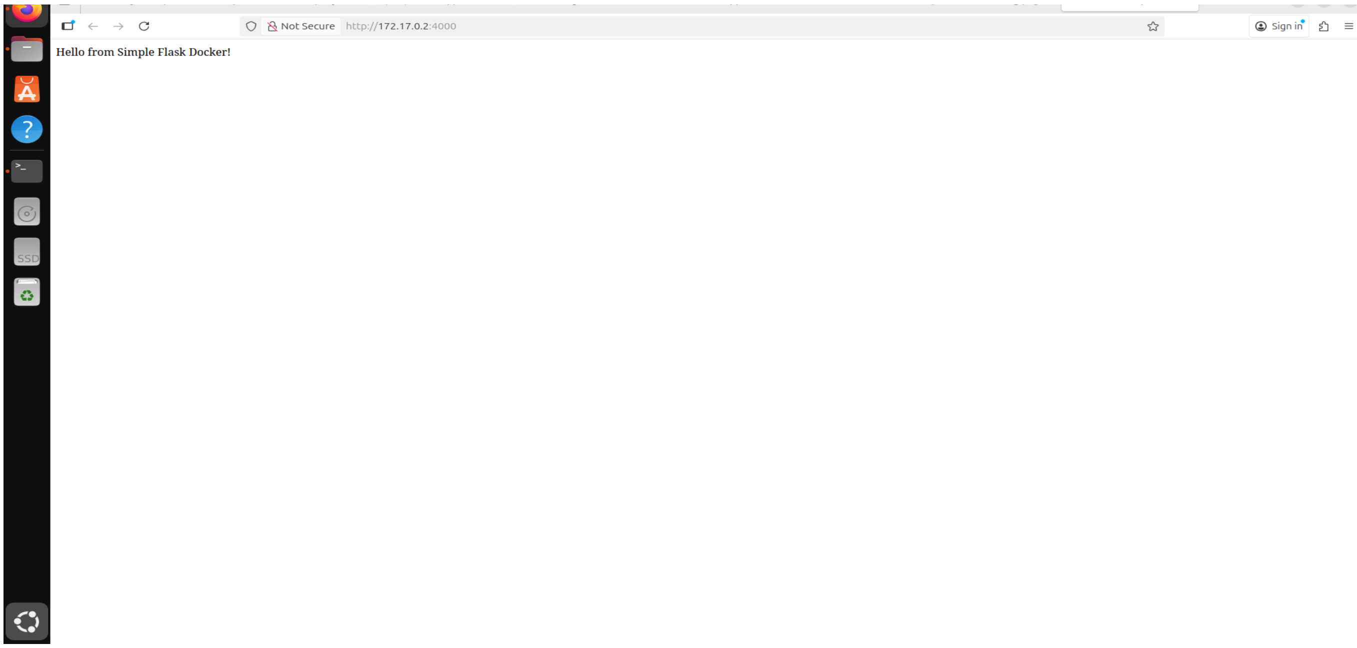
Step 7: Test the Application

Open browser and go to:

http://localhost:5000

Output displayed:

Hello from Simple Flask Docker!


Not Secure
http://172.17.0.2:4000
Sign in
Hello from Simple Flask Docker!